SCRDR for Real-World Classification

လက်တွေ့ပြဿနာများတွင် SCRDR ကိုအသုံးပြုခြင်း

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Htwe Myat Cho, Yee Mon Thant, Kaung Pyae Min Thein, Kaung Htet Htun, Hnin Yadana Lwin
Min Bhone San, Aung Hein, Mya Moe Wai, Thet Su Sann, Zwe Wint Ti

Mental Health

Banking Marketing

What is SCRDR?

SCRDR ဆိုတာဘာလဲ?

Rule-Based Learning

Rule-based
သင်ယူမှု

Human expert writes IF→THEN rules. Exceptions are added on top of existing rules — no rule is ever deleted.

Ripple-Down Structure

Exception tree
ဖွဲ့စည်းမှု

Each mistake triggers a new exception rule underneath the failing rule, forming a decision tree.

Single Class Output

တစ်မျိုးတည်း
ခွဲခြားမှု

SCRDR assigns exactly ONE class per case — perfect for binary problems like depressed / not depressed.

How the Interactive Build Works

Interactive Build ဘယ်လိုအလုပ်လုပ်သလဲ

1. **System shows a misclassified case**
   မှားယွင်းသောကိစ္စတစ်ခု ပြသသည်

2. **Human expert picks a feature + condition**
   လူသားကျွမ်းကျင်သူ feature နှင့် condition ရွေးသည်

3. **Rule is added to the exception tree**
   Rule ကို exception tree ထဲထည့်သည်

4. **Repeat until all training cases are correct**
   Training case အားလုံးမှန်သည်အထိ ထပ်လုပ်သည်

⚠️ 100% train accuracy is expected — RDR patches every training case by design. Test accuracy is what really matters.

Our Two Classification Problems

ကျွန်တော်တို့ရွေးချယ်သော ပြဿနာနှစ်ခု

🧠 Mental Health

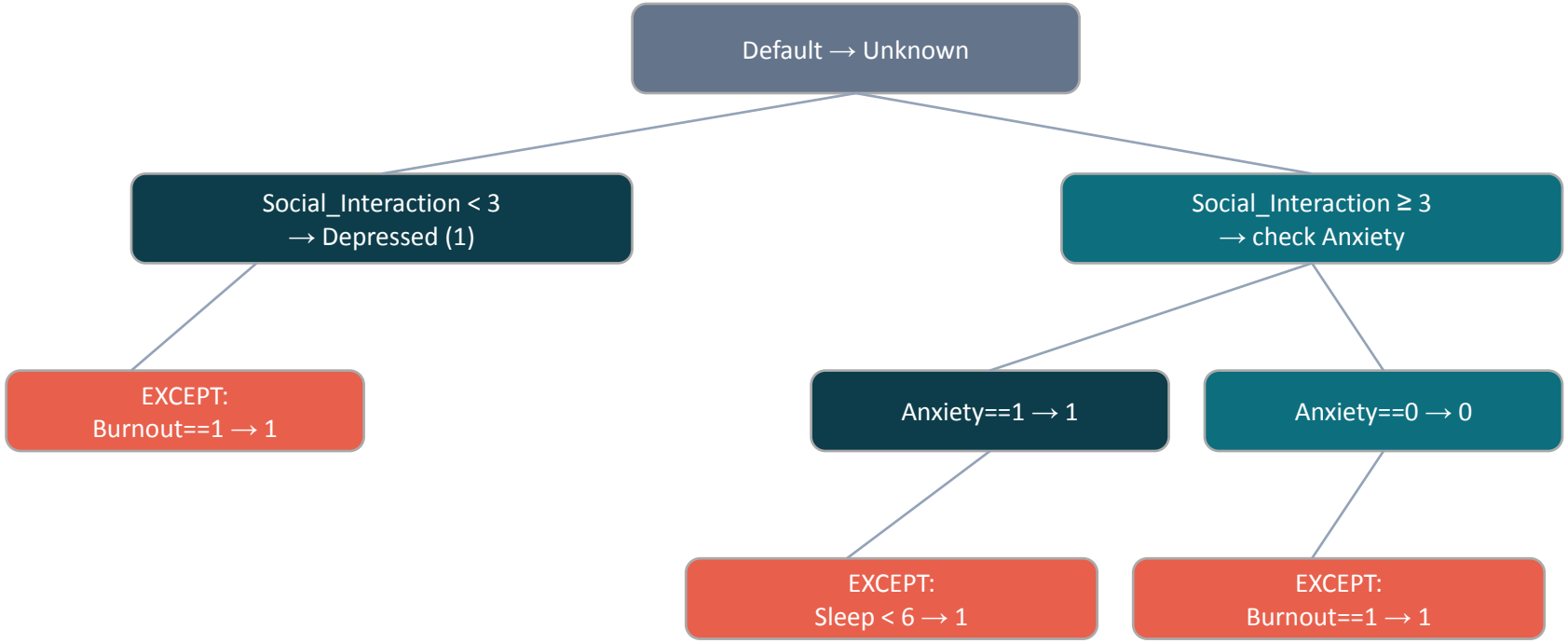
Target	Depression (0 / 1)
Source	Kaggle – Mental Health Dataset
Train	50 rows (25 / 25 balanced)
Test	20 rows (10 / 10 balanced)
Features	17 (age, sleep, anxiety, burnout…)
Rules built	Human expert – interactive session

🏦 Banking Marketing

Target	deposit (yes / no)
Source	Kaggle – Bank Marketing Dataset
Train	50 rows (balanced split)
Test	20 rows (balanced split)
Features	16 (Age, job, balance, duration…)
Rules built	Human expert – interactive session

SCRDR Rule Tree – Mental Health

Mental Health Rule Tree ဖွဲ့စည်းပုံ

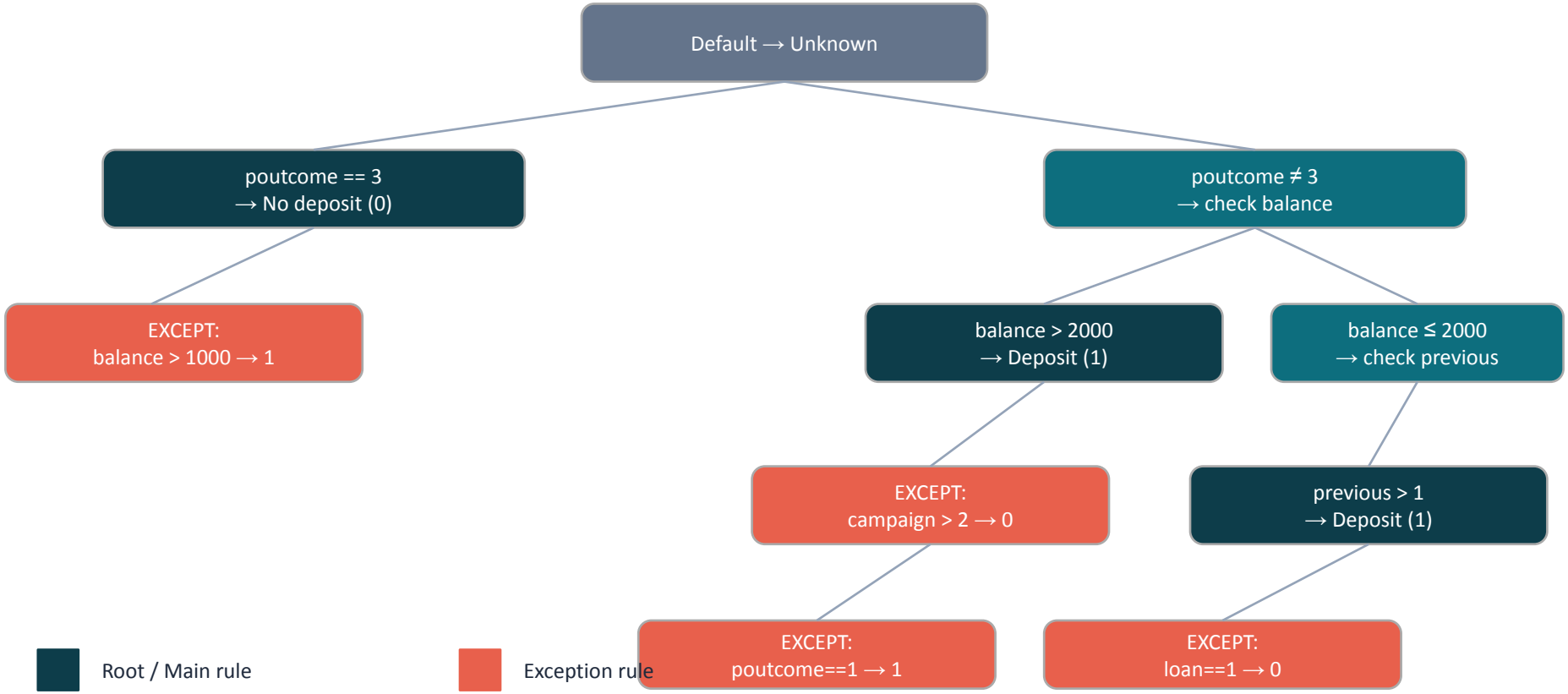


Root / Main rule

Exception rule

SCRDR Rule Tree – Banking Marketing

Banking Rule Tree ဖွဲ့စည်းပုံ



Results – Mental Health

Mental Health ရလဒ်များ

1.00

Train F1

လေ့ကျင့်မှု

0.55

Test F1 (SCRDR)

စမ်းသပ်မှု

55%

Test Accuracy

တိကျမှု

Test Classification Report (2000 rows)

Class	Precision	Recall	F1
0 (Not Depressed)	0.54	0.65	0.59
1 (Depressed)	0.56	0.45	0.50
Weighted Avg	0.55	0.55	0.55

vs Other Methods

Method	F1
Interactive SCRDR	0.55
Auto SCRDR	0.54
Random Forest	**0.58** ⬆️

Results – Banking Marketing

Banking Marketing ရလဒ်များ

0.96

Train F1

လေ့ကျင့်မှု

0.50

Test F1 (SCRDR)

စမ်းသပ်မှု

50%

Test Accuracy

တိကျမှု

Test Classification Report (11162 rows)

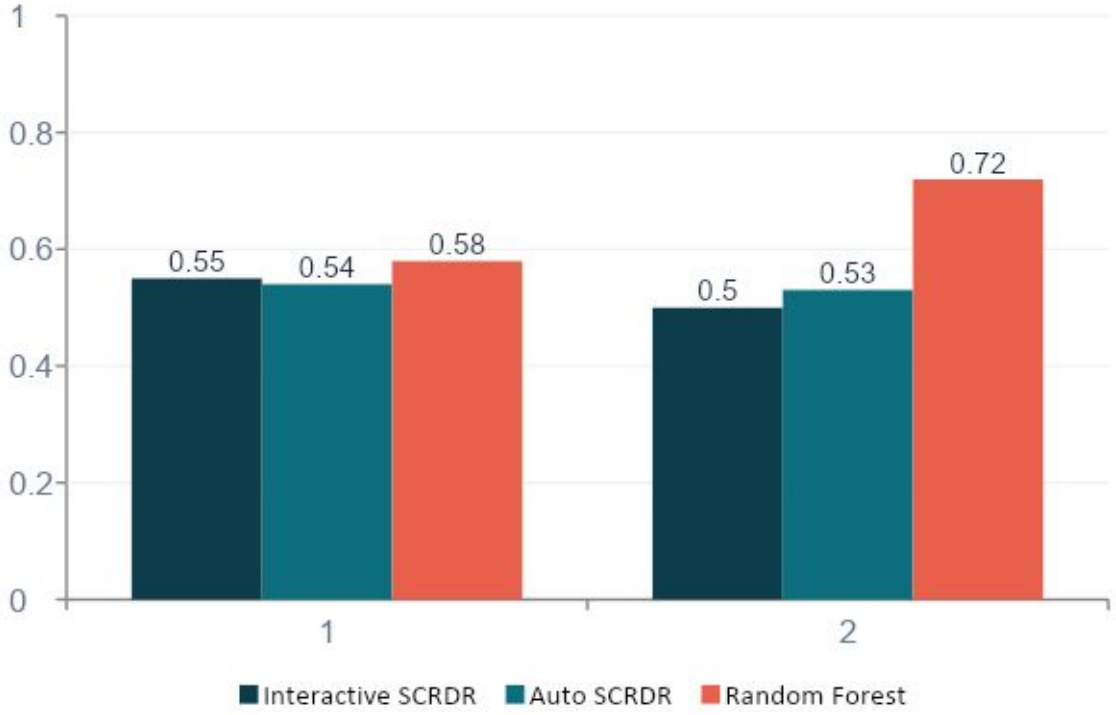
Class	Precision	Recall	F1
0 (No deposit)	0.54	0.38	0.45
1 (Deposit)	0.48	0.64	0.55
Weighted Avg	0.51	0.51	0.50

vs Other Methods

Method	F1
Interactive SCRDR	0.50
Auto SCRDR	0.53
Random Forest	0.72 ⬆️

Model Comparison

မော်ဒယ်များ နှိုင်းယှဉ်ချက်



Test F1 Summary

Method	MH	Bank
Interactive SCRDR	0.55	0.50
Auto SCRDR	0.54	0.53
Random Forest	**0.58** ⬆️	**0.72** ⬆️

Takeaway

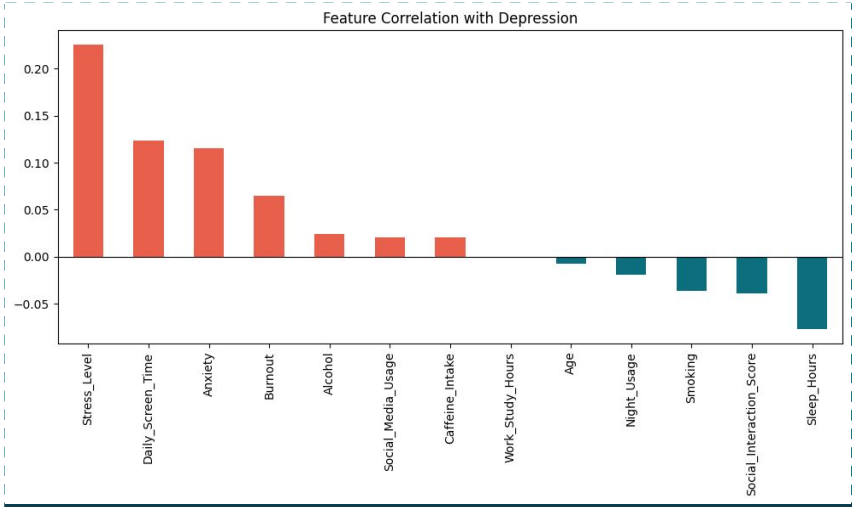
Mental Health: SCRDR nearly matches RF — clear domain signals help.

Banking: RF wins by a large margin (0.72 vs 0.50) — complex feature interactions favour ML over rules.

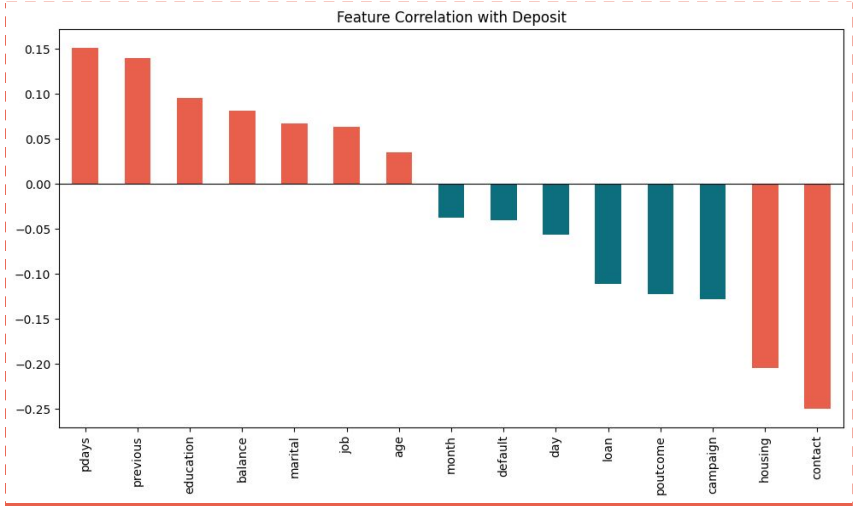
Feature Importance — What the Data Says

Data က ပြောတဲ့ Feature အရေးပါမှု

💡 **Rules that focus on highly correlated features generalise better — regardless of how many rules are built or how much data was used to train.**



Mental Health — Top: Stress_Level (0.23), Anxiety (0.12)



Banking — Top: pdays (0.15), previous (0.14), contact (-0.25)

Key Findings

အဓိကတွေ့ရှိချက်များ

🎯

RDR trains to 100% — by design

RDR သည် training data ကို 100% မှန်ကန်အောင် တည်ဆောက်သည် — Each exception patch fixes one case without breaking previous rules. This always results in perfect training accuracy.

⚠️

High train ≠ high test

Training မှုကောင်းသော်လည်း Test မှုကောင်းချင်မှ ကောင်းသည် — Mental Health: 1.00 → 0.55. Banking: 0.96 → 0.40. The gap reveals overfitting to training cases.

🧑‍💼

Human expertise is the key ingredient

လူသားကျွမ်းကျင်မှုသည် အဓိကအချက်ဖြစ်သည် — Automated rule-building (scripted logic) gave 0.30 F1. Human-guided building gave 0.55 — showing that genuine reasoning matters.

📊

Feature choice determines generalisation

Feature ရွေးချယ်မှုသည် generalisation ကို သတ်မှတ်သည် — Teammate's 3 rules using Stress_Level (corr 0.23) → 0.59 F1. Your rules using Social_Interaction_Score (corr ~0.00) → 0.55 F1. Correlation matrix doesn't lie.

Conclusion

နိဂုံးချုပ်

1. **SCRDR is transparent and explainable — every decision traces back to a human-written rule.**

   SCRDR သည် ဆုံးဖြတ်ချက်တိုင်းကို လူသားရေးသားသော rule မှ ဆွဲထုတ်နိုင်သည်

2. **It works best when domain signals are clear and the expert genuinely reasons per case.**

   Domain signal ရှင်းလင်းပြီး ကျွမ်းကျင်သူ တကယ် အသုံးချမှသာ ကောင်းမွန်သောရလဒ်ရသည်

3. **More data and careful rule design can reduce the train/test gap significantly.**

   Data ပိုများပြီး rule ရေးဆွဲမှု ဂရုစိုက်ပါက Train/Test ကွာဟချက် လျော့ကျနိုင်သည်

4. **SCRDR complements ML — use it where interpretability matters most.**

   SCRDR သည် ML ကို ဖြည့်ဆည်းသည် — ရှင်းလင်းပြတ်သားမှု အရေးကြီးသောနေရာတွင် သုံးပါ

References

1. https://www.kaggle.com/datasets/jajidhasan/mental-health

2. https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset

3. Compton, P., & Kang, B. H. (2021). Ripple-Down Rules: The Alternative to Machine Learning. CRC Press.